

Anita Petrovic

[LinkedIn](#) | Geneva, Switzerland | Canadian & Croatian (EU) citizen

Work Experience

CERN, Mechanical Engineer

September 2024 - Present

Mechanical engineer supporting the [FCC project](#) within CERN's cryogenics department

Geneva, Switzerland

- Support cryogenic engineering for superconducting radio-frequency (RF) cryomodules, transfer lines, and surface and underground technical infrastructure.
- Sized helium two-phase line and rupture-disk line concepts for RF cryomodules to control vapour velocity, helium inventory and level stability, flow velocity, and pressure differentials.
- Develop Python models for predicting cryogenic distribution heat loads and properties, including conduction/radiation, pressure drop, temperature/enthalpy evolution, mass-flow profiles, JT/flash behaviour, node mixing, and distributed heat inleaks.
- Define cryogenic requirements for utilities, electrical racks, building interconnections, installation strategy, and integration constraints.
- Coordinate cryogenic design and integration activities with multidisciplinary engineering and infrastructure teams; serve as the CAD/PLM point of contact for cryogenics.
- Prepare technical inputs, reports, and presentations for major feasibility studies, technical reports, and international conferences.

Lucid Motors, Chassis Mechanical Design Engineering Intern

May 2022 - August 2022

Luxury Electric Vehicle Company

Newark, CA

- Owned the design of a bushing fixture to test components of the Lucid Air suspension system.
- Used CATIA FEA to limit fixture deflection and verified that the design could withstand a maximum load of 18.2 kN.
- Created and executed a static and dynamic testing plan with the CTW LA-48 actuator.
- Wrote a MATLAB program to analyze test data from suspension bushings to verify static and dynamic stiffness performance.
- Designed, prototyped, and tested string potentiometer fixtures to collect and analyze rear steering data.
- The projects involved setting up involved setting up sensors and DAQ. VBOX and CANalyzer were used for data logging.
- Developed a concept jig model for future rear steering testing of the Lucid Gravity SUV.

Tesla, Structural Design Engineering Intern

September 2021 - December 2021

Electric Vehicle and Clean Energy Company

Fremont, CA

- Owned the redesign of SEMI structural roof components, achieving a 5.4% mass reduction while addressing stamping feasibility, mass manufacturing, durability, and component integration.
- Independently designed and implemented new toe features for Model S/X, reducing cost and weight by over 7% while retaining structural integrity for frontal and corner crashes.
- Iterated designs for FDM-printed 3D fixtures with feedback from factory technicians that addressed NVH issues and reducing application time of adhesives by 10.5%.
- Updated GD&T for new product designs for Model S/X.

Projects

EcoCar, Mechanical Team Lead

September 2018 - May 2023

Shell Eco-marathon, Fuel Cell Vehicle Team - University of Alberta

Edmonton, AB

- Led mechanical work for a fuel-cell vehicle team, managing 12 members, scheduling, sponsorships, and a \$25,000 annual budget.
- Modeled the fuel cell, supercapacitor, DC/DC converter, and motor system in Simulink; designed a 2020 Urban-Concept wheel hub that achieved a 48% weight reduction.
- Managed a four-month fiberglass and carbon composite body layup process.
- Developed compact suspension/steering concepts for a confined monocoque.

Education

University of Alberta

June 2023

BSc Mechanical Engineering, GPA: 3.9/4.0

Edmonton, AB

- Scovan Engineering Prize in Design Project Management

Skills

CAD: CATIA V5/3DX, SOLIDWORKS (CSWP certification)

Analysis/Programming: Python, Matplotlib, MATLAB, ANSYS, CATIA FEA, Excel

Engineering: Cryogenic systems, heat-transfer problems, pressure-drop calculations, design for manufacturing, mechanical design, composite manufacturing, technical reporting